

WASTE FOOD MANAGEMENT SYSTEM



Project ID - G02

Project Advisor: Prof. Hamza Masood

Submitted By

MUHAMMAD USMAN ALI **bsf2108831**

REHMAN ZAHOR **bsf2108808**

ABBAS AHMED MIR **bsf2108821**

University of Education

WASTE FOOD MANAGEMENT SYSTEM

BS IN INFORMATION TECHNOLOGY-2025

A project submitted in partial fulfillment of the requirements for
the award of the degree of BS in information Technology

BS IN INFORMATION TECHNOLOGY / GUJARKHAN CAMPUS

UNIVERSITY OF EDUCATION

LAHORE

September 2025

DECLARATION

I declare that this project title entitled “**Waste Food Management System**” is the result of my own research and development except as cited in the references. This project has not been accepted for any degree and is not concurrently submitted in candidate for any other degree. At any time if my statement is found to be incorrect even afterwards of BS in Information Technology, the university has the right to withdraw my BS in Information Technology degree.

Signature: _____

Signature: _____

Name: MUHAMMAD USMAN ALI

Name: REHMAN ZAHOOR

Date: _____

Date: _____

Signature: _____

Name: ABBAS AHMED MIR

Date: _____

CERTIFICATE OF APPROVAL

This is to certify that the project work presented in this documentation entitled, **“Waste Food Management System”**, was conducted by **“Muhammad Usman Ali”**, **“Rehman Zahoor”**, **“Abbas Ahmed Mir”** under the supervision of **“Prof. Hamza Masood”**. No part of this project has been submitted anywhere else for any degree. This project is submitted to the “University of Education” in partial fulfillment of the requirements of the degree of BS in Information Technology.

Signature: _____

Signature: _____

Name: MUHAMMAD USMAN ALI

Name: REHMAN ZAHOOR

Date: _____

Date: _____

Signature: _____

Name: ABBAS AHMED MIR

Date: _____

ABSTRACT

Background:

Food wastage is a growing global issue, with hotels and restaurants discarding large quantities daily while many communities suffer from hunger. Efficient food redistribution can reduce waste and support underprivileged communities.

Objectives/Aim of the Study:

The aim of this project is to design and develop a web-based platform that connects restaurants and hotels (donors) with NGOs (receivers) to donate surplus food safely and efficiently. The system seeks to minimize food wastage and facilitate timely food redistribution.

Research Methodologies:

The project was implemented using the MERN stack (MongoDB, Express.js, React.js, Node.js). JWT authentication was integrated to ensure secure access with role-based dashboards for Admin, Restaurant, and NGO users. The platform enables restaurants to list donations, NGOs to claim them, and admins to oversee the process. Data flow and role-based operations were structured using an API-driven architecture.

Findings:

The developed system successfully facilitates the donation and claiming process in real time. Features such as donation lifecycle management (active, expired, picked up, delivered) and retention of completed/expired records enhance usability. The platform also supports scalability and responsiveness, ensuring smooth user interaction across devices.

Conclusions:

The project demonstrates that a centralized food donation platform can significantly reduce waste while assisting NGOs in addressing food insecurity. The adoption of modern web technologies ensures performance, security, and maintainability.

Implications:

This system can be extended to larger communities and integrated with mapping and notification services to improve efficiency. It has potential to be deployed at scale, encouraging sustainable practices in the hospitality industry.

Limitations:

The project was limited to web-based functionality with basic map integration planned. Features such as real-time delivery tracking, mobile applications, and government regulatory compliance were outside the scope of this phase. Future improvements can address these limitations for wider adoption.

1 Chapter No. 1: Gathering & Analyzing Information

1.1 Introduction

Food waste is a major challenge faced by the hospitality industry worldwide, with restaurants and hotels discarding large quantities of edible food daily. At the same time, millions of people remain food insecure. Bridging the gap between surplus food and communities in need requires an efficient, reliable, and transparent system. This project addresses this issue by developing a Food Waste Donation Platform using the MERN stack. The platform connects restaurants (donors) with NGOs (receivers) to ensure surplus food is donated and distributed safely, thereby reducing waste and contributing to social welfare.

1.2 Problem Statement

Despite the availability of surplus food in restaurants and hotels, there is no structured system to channel it to NGOs efficiently. Current practices rely on manual communication, which is time-consuming, unreliable, and lacks transparency. As a result, food wastage continues, and underprivileged communities miss the opportunity to benefit. A centralized web-based solution is needed to automate, secure, and streamline this process.

1.3 Goal & Objectives

1.3.1 Goal:

To design and develop a web-based platform that minimizes food waste by enabling efficient donation and redistribution of surplus food from restaurants to NGOs.

1.3.2 Objectives:

- To build a role-based system with dashboards for Admin, Restaurant, and NGO.
- To implement secure authentication using JWT for safe access and data protection.
- To allow restaurants to donate surplus food and NGOs to claim it in real-time.
- To maintain donation records including active, expired, and completed donations.
- To provide an admin panel for monitoring, managing, and generating insights.
- To integrate map-based features to improve accessibility and visibility of donation points.

1.4 Research Questions

1. How can technology reduce food wastage in the hospitality industry?
2. What system features are required to ensure transparent and efficient food donation?
3. How can role-based access improve the usability and security of the system?
4. What methodology and technology stack best supports scalability and performance in such platforms?

1.5 Methodology

Available Methodologies

- **Waterfall Model:** A linear approach where phases are completed sequentially. Suitable for small projects but less flexible to changes.
- **Agile Methodology:** An iterative and incremental model that focuses on adaptability, collaboration, and continuous delivery.
- **Spiral Model:** Combines iterative development with risk analysis, but is complex and resource-intensive.

1.6 Chosen Methodology

- The Agile Methodology was chosen for this project. Development was carried out in iterative sprints, allowing continuous testing, feedback, and feature enhancement.
- Reasons for Chosen Methodology
- Flexibility to adapt to changing requirements.
- Continuous integration and testing ensure high-quality outcomes.
- Incremental development improves risk management.
- Better collaboration between frontend and backend development phases.

1.7 Definitions, Acronyms, and Abbreviations

- MERN: MongoDB, Express.js, React.js, Node.js
- JWT: JSON Web Token, used for secure authentication
- NGO: Non-Governmental Organization
- Admin: Role responsible for monitoring and managing the system
- Donor: A restaurant or hotel contributing surplus food
- Receiver: An NGO that claims and distributes food
- API: Application Programming Interface, enabling communication between frontend and backend
- CRUD: Create, Read, Update, Delete – basic database operations

2 Chapter No 2: Software Requirement Specification

2.1 Stakeholders Characteristics

- **Restaurants:**
Entities that donate surplus food via the platform. They expect a fast, secure, and simple system to post available food, track donation history, and receive notifications. They require a dashboard tailored to their role.
- **NGOs:**
Organizations that receive and distribute the donated food to those in need. They expect real-time updates on food availability, an easy interface to request or claim donations, and access to location-based features (e.g., maps for pickup coordination).
- **Admin:**
Admins oversee the entire system, including NGO and restaurant user accounts, food donation logs, reports, and analytics. They expect a powerful dashboard with full control, role-based access, and moderation capabilities.
- **End Users (Beneficiaries - Indirect Stakeholders):**
Although not directly using the system, they benefit from the food delivered by NGOs. Their satisfaction reflects the overall success of the platform's mission.
- **Developers & Testers:**
Responsible for building, debugging, and testing the MERN stack-based application. They expect structured modules, consistent code practices, proper documentation, and accessible testing environments.
- **Supervisors & Evaluators:**
Academic stakeholders who assess the technical and social impact of the project. They expect a polished, functional, well-documented, and ethically sound system aligned with final year project standards.

2.2 Domain Requirements

- **Food Waste Donation System:**
A digital platform that facilitates the donation of surplus food from restaurants to NGOs that distribute it to those in need.
- **User Authentication:**
Secure user registration and login system using JWT, with session management for Restaurants, NGOs, and Admins.
- **Food Listing and Requesting:**
Restaurants can list available surplus food, including quantity, type, and expiry; NGOs can browse, filter, and request food donations.
- **Map Integration and Location Tracking:**
NGOs can view available food donations on a map, based on proximity, using location-based services for efficient pickup coordination.
- **Role-based Access Control:**
System functionalities differ by role — Restaurants manage donations, NGOs claim food, and Admins monitor and manage all users and data.
- **Data Management:**
All user accounts, food donations, request logs, pickup statuses, and system feedback are securely stored and managed in the MongoDB database.
- **Report Generation:**
Admins can generate reports on donation volumes, NGO activity, restaurant contributions, and overall system performance.
- **Feedback System:**
NGOs and Restaurants can submit feedback or concerns; Admins can review and respond to improve service quality.

2.3 Functional Requirements:

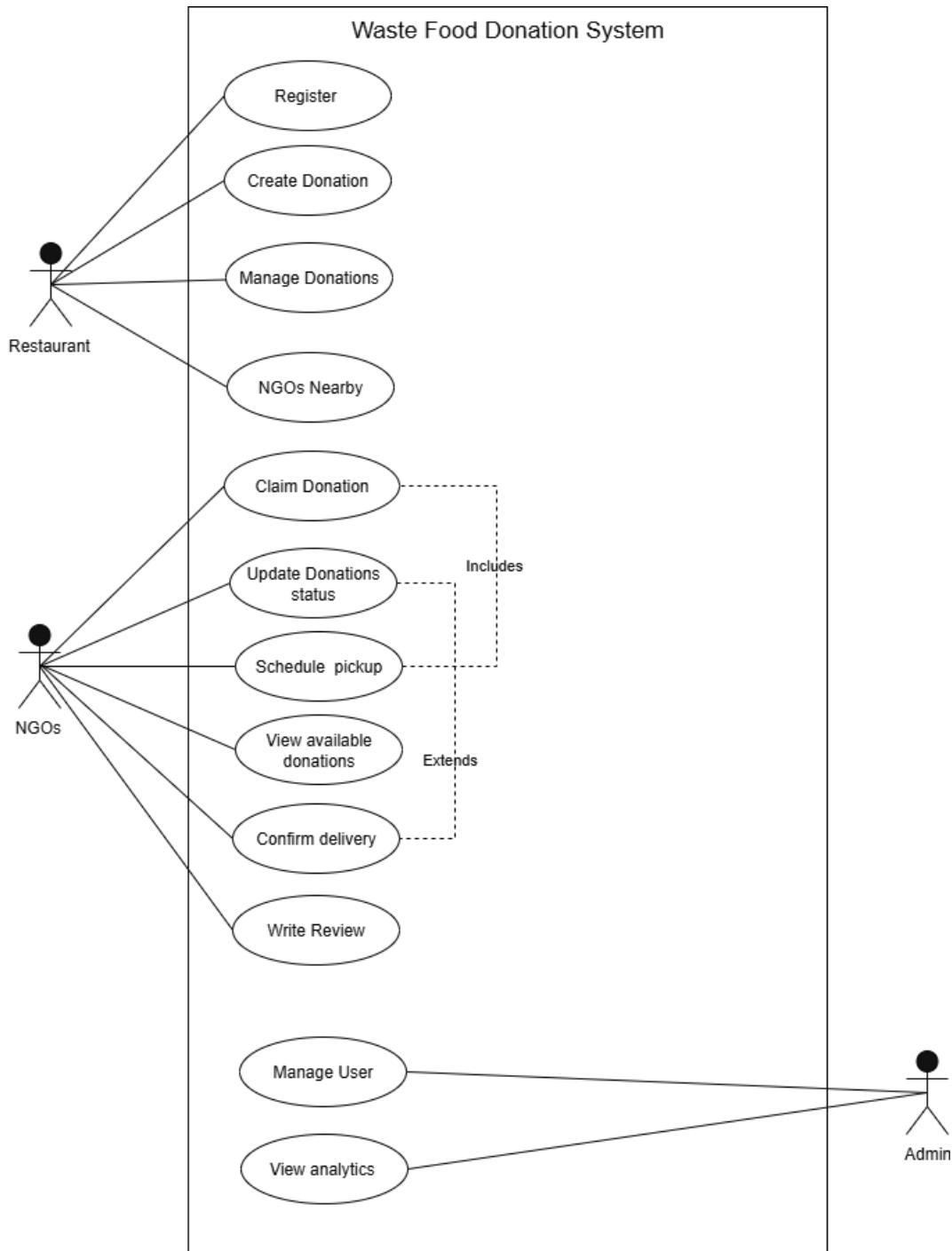
No.	Requirements Title	Requirements Description
FR1	Register as Restaurant/NGO	Restaurants/NGOs register by submitting details; system emails login credentials.
FR2	Post Food Donation	Logged-in restaurants can post food donation details (quantity, location, expiry).
FR3	Claim Food Donation	NGOs can browse and claim donations; system notifies the restaurant.
FR4	Provide Feedback	Users submit feedback about the system; stored for admin review.
FR5	View Claimed Donations	NGOs can track their claimed donations and pickup status.
FR6	Approve NGO/Restaurant	Admin approves/rejects registrations; system updates status and notifies.
FR7	Send Notification	System sends automatic alerts for donation status changes (e.g., claimed, ready).
FR8	View Reports	Admin can generate reports (donations, claims, registrations).

2.4 Non-Functional Requirements:

No.	Requirement Title	Requirement Description
NFR1	Security	The system will ensure secure communication and data handling (JWT, encryption).
NFR2	Usability and Accessibility	The UI will be user-friendly, responsive, and accessible.
NFR3	Scalability / Performance	The system should support 50+ concurrent users.
NFR4	Access Control and Auditability	All access will be role-restricted and logged.
NFR5	Performance (Response Time)	Reports will be generated in under 30 seconds.
NFR6	Maintainability / Documentation	Online documentation will guide users and admins.

3 Chapter No 3: Analysis [Use Case Description and Use Case Model]

3.1 Use Case Model



3.2 Use Case Description

3.2.1 UC Number: 01

ID	UC-01
Use case Name	Register
Actor	Restaurant, NGO
Description	Register as NGO or restaurant
Precondition	The system is aligned and accessible
Postcondition	User can login with registered credentials
Priority	High

3.2.2 UC Number: 02

ID	UC-02
Use case Name	Create Donation
Actor	Restaurant
Description	Create a new food donation with details
Precondition	User is logged in as restaurant
Postcondition	New donation is created and visible to NGOs
Priority	High

3.2.3 UC Number: 03

ID	UC-03
Use case Name	Manage Donation
Actor	Restaurant
Description	View, edit or delete donation listing
Precondition	User is logged in as restaurant
Postcondition	Donations are updated or removed from system
Priority	High

3.2.4 UC Number: 04

ID	UC-04
Use case Name	NGOs nearby
Actor	Restaurant
Description	View nearby NGOs
Precondition	User is logged in as restaurant
Postcondition	Nearby NGOs are displayed
Priority	Medium

3.2.5 UC Number: 05

ID	UC-05
Use case Name	View Available Donation
Actor	NGOs
Description	Browse available donations
Precondition	User is logged in as NGOs
Postcondition	Available donations are displayed to NGO
Priority	High

3.2.6 UC Number: 06

ID	UC-06
Use case Name	Claim Donation
Actor	NGO
Description	Claim an available donation for pickup
Precondition	User is logged in as an NGO
Postcondition	Donations status changes to claimed
Priority	High

3.2.7 UC Number: 07

ID	UC-07
Use case Name	Schedule Pickup
Actor	NGO
Description	Arrange pickup time with restaurant for claim donation
Precondition	Donation has been claimed by NGO
Postcondition	Pickup is scheduled
Priority	High

3.2.8 UC Number: 08

ID	UC-08
Use case Name	Write Reviews
Actor	NGO
Description	Review donation quality after receiving
Precondition	NGO has received the donation
Postcondition	Review added
Priority	Medium

3.2.9 UC Number: 09

ID	UC-09
Use case Name	Manage users
Actor	Admin
Description	Manage user accounts including blocking
Precondition	User is logged in as admin
Postcondition	Users accounts are updated
Priority	Medium

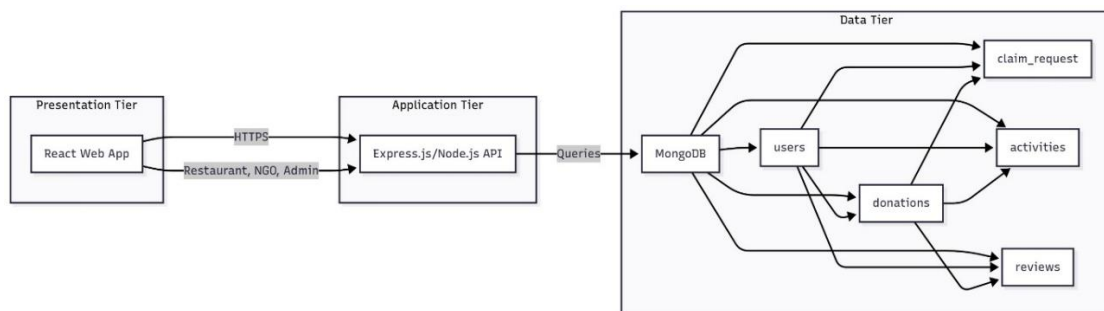
3.2.10 UC Number: 10

ID	UC-10
Use case Name	View analytics
Actor	Admin
Description	View system analytics and reports
Precondition	User is logged in as admin
Postcondition	Analytics data is displayed to admin
Priority	Low

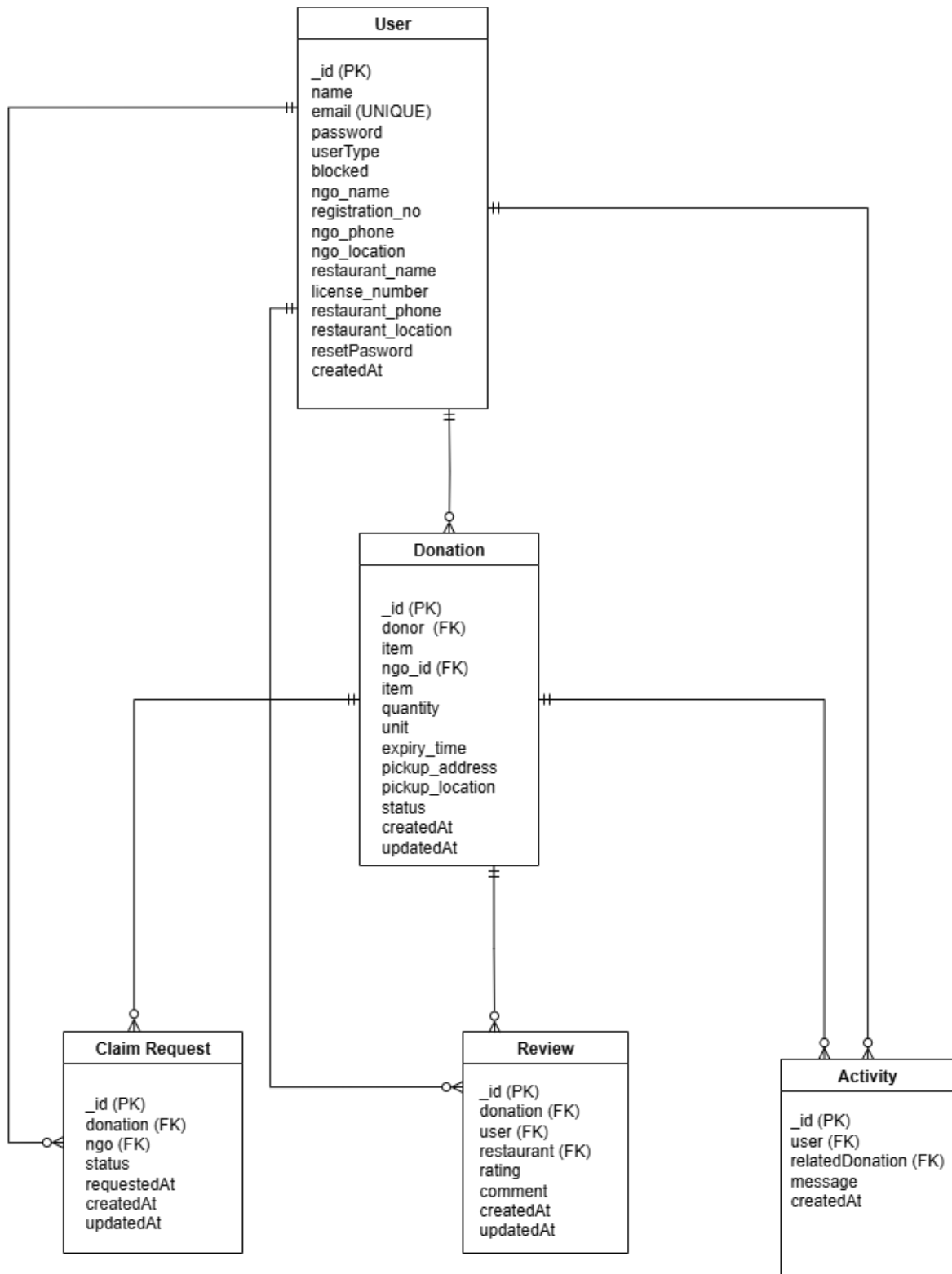
4 Chapter No 4: Design

4.1 Architecture Diagram

The system follows a three-tier architecture built on the MERN stack. The frontend (React.js) provides user interfaces for Admin, NGO, and Restaurant roles. The backend (Node.js + Express.js) manages business logic, authentication, and role-based operations. The database (MongoDB) stores user information, donations, claims, and transaction history. Communication between frontend and backend occurs via RESTful APIs secured with JWT authentication. This architecture ensures scalability, modularity, and security.



4.2 Entity Relationship Diagram (ERD – Description)



The system uses **MongoDB** with five main collections:

4.2.1 Users

- Stores details of Restaurants, NGOs, and Admins.
- **Relations:**
 - One Restaurant → many Donations.
 - One NGO → many Claim Requests & Reviews.
 - Users generate many Activities.

4.2.2 Donations

- Food donations created by Restaurants.
- **Relations:**
 - Belongs to one Restaurant (donor).
 - Can be claimed by one NGO.
 - Can have many Claim Requests.
 - Can have one Review after completion.
 - Generates Activities (created, claimed, delivered).

4.2.3 Claim Requests

- NGOs request to claim donations.
- **Relations:**
 - Linked to one Donation.
 - Made by one NGO.
 - Generates Activities when updated.

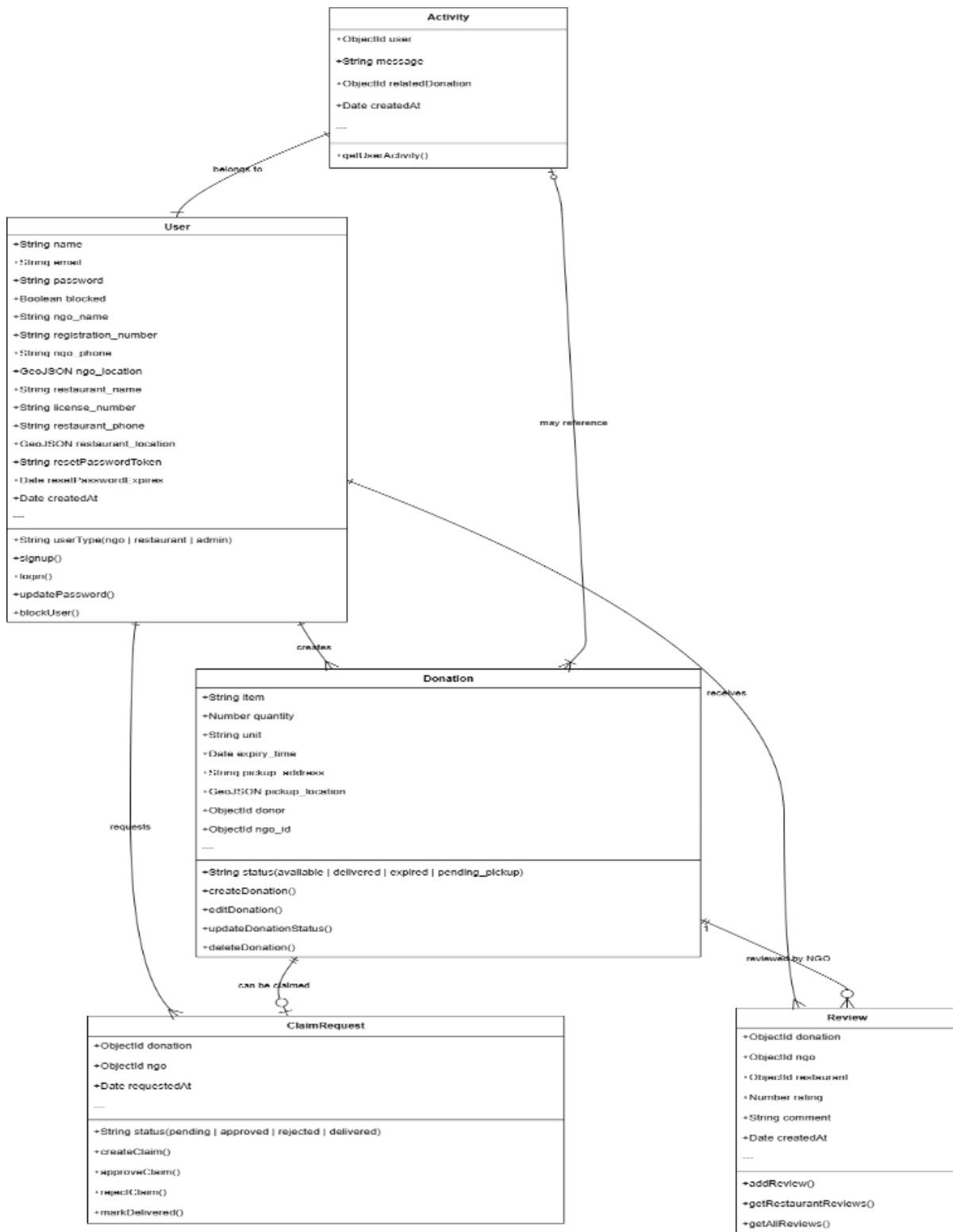
4.2.4 Reviews

- NGOs give feedback on Restaurants after receiving food.
- **Relations:**
 - One Review per completed Donation.
 - Written by one NGO, for one Restaurant.

4.2.5 Activities

- Logs actions and notifications.
- **Relations:**
 - Belongs to one User.
 - May reference a Donation.

4.3 Class Diagram



The class diagram represents the main entities of the Food Waste Donation System, their attributes, behaviors, and relationships:

4.3.1 User

- Represents system users (Restaurant, NGO, or Admin).
- Attributes include login credentials, organization details, location, and status (blocked/unblocked).
- Key operations: signup, login, password management, and account control.
- **Relations:** A User can create multiple Donations, make Claim Requests, write Reviews, and generate Activities.

4.3.2 Donation

- Represents a food item donated by a Restaurant.
- Stores item details, quantity, expiry, pickup info, status, and references to donor/NGO.
- Key operations: create, edit, update status, and delete.
- **Relations:** A Donation may have one Claim Request and multiple Reviews, and it can generate Activities.

4.3.3 ClaimRequest

- Represents an NGO's request to claim a Donation.
- Includes request time and status (pending, approved, rejected, delivered).
- Key operations: create, approve, reject, mark delivered.
- **Relations:** Each ClaimRequest is linked to one Donation and one NGO (User).

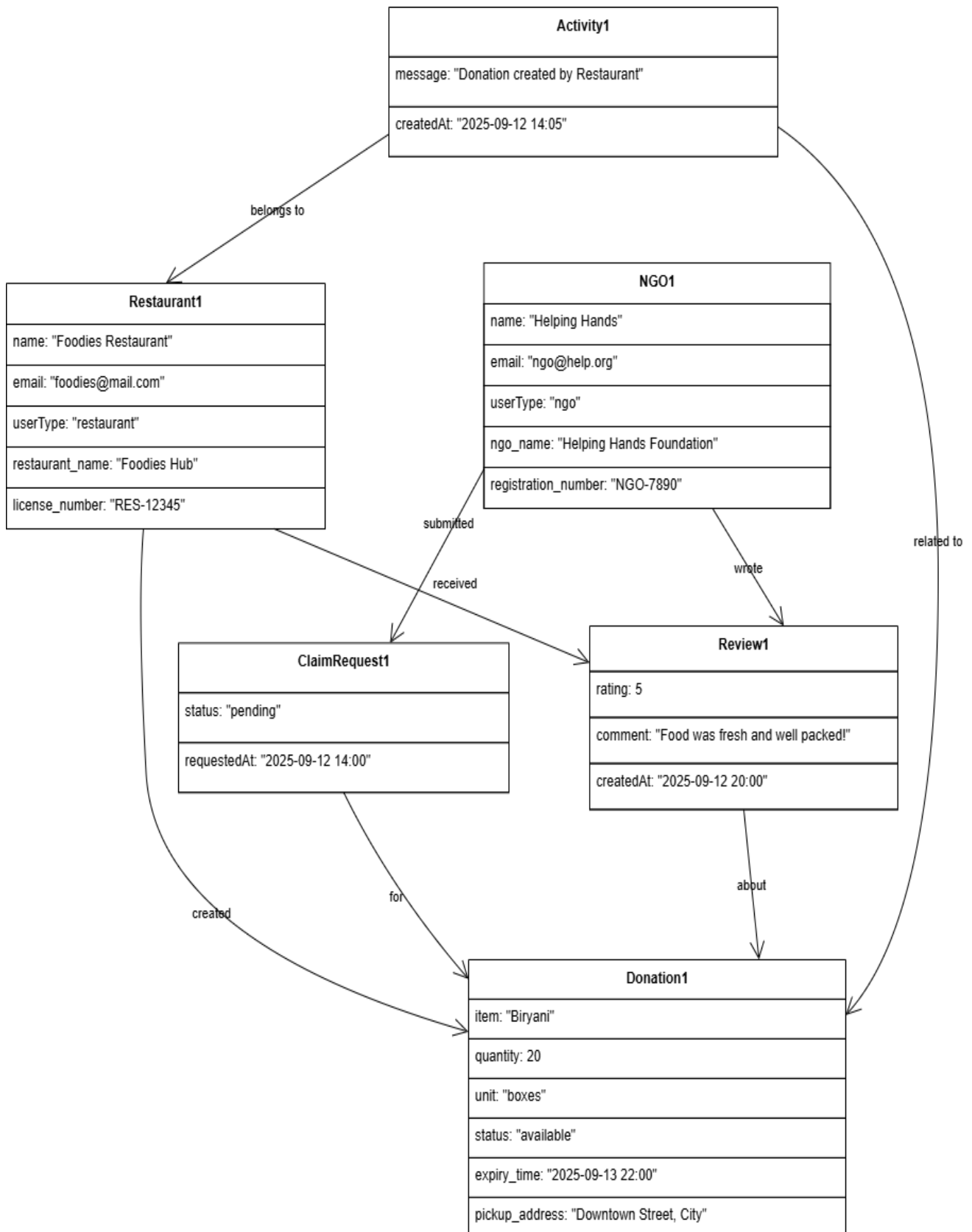
4.3.4 Review

- Represents NGO feedback on Restaurants after donation completion.
- Stores rating, comments, and links to related Donation, NGO, and Restaurant.
- Key operations: add review, fetch restaurant reviews, fetch all reviews.
- **Relations:** Each Review is tied to a Donation and Users (NGO and Restaurant).

4.3.5 Activity

- Represents system activity logs and notifications.
- Stores user actions, related donation, and timestamp.
- Key operation: retrieve user activity.
- **Relations:** Each Activity belongs to a User and may reference a Donation.

4.4 Object Diagram



The object diagram represents a **real-world snapshot** of the Food Waste Donation System, showing specific instances of users, donations, requests, reviews, and activities, along with their relationships:

4.4.1 Restaurant1 (Foodies Restaurant)

- A restaurant user with license number RES-12345.
- **Creates** a donation of food (Biryani, 20 boxes).

4.4.2 NGO1 (Helping Hands Foundation)

- An NGO user with registration number NGO-7890.
- **Submits** a claim request for the donation.
- **Writes** a review about the donation after receiving it.

4.4.3 Donation1 (Biryani, 20 boxes)

- Posted by Restaurant1.
- **Claimed by** NGO1 through ClaimRequest1.
- Later **reviewed** by NGO1 (Review1).

4.4.4 ClaimRequest1

- Status: pending, requested on 2025-09-12 14:00.
- Represents NGO1's attempt to claim Donation1.

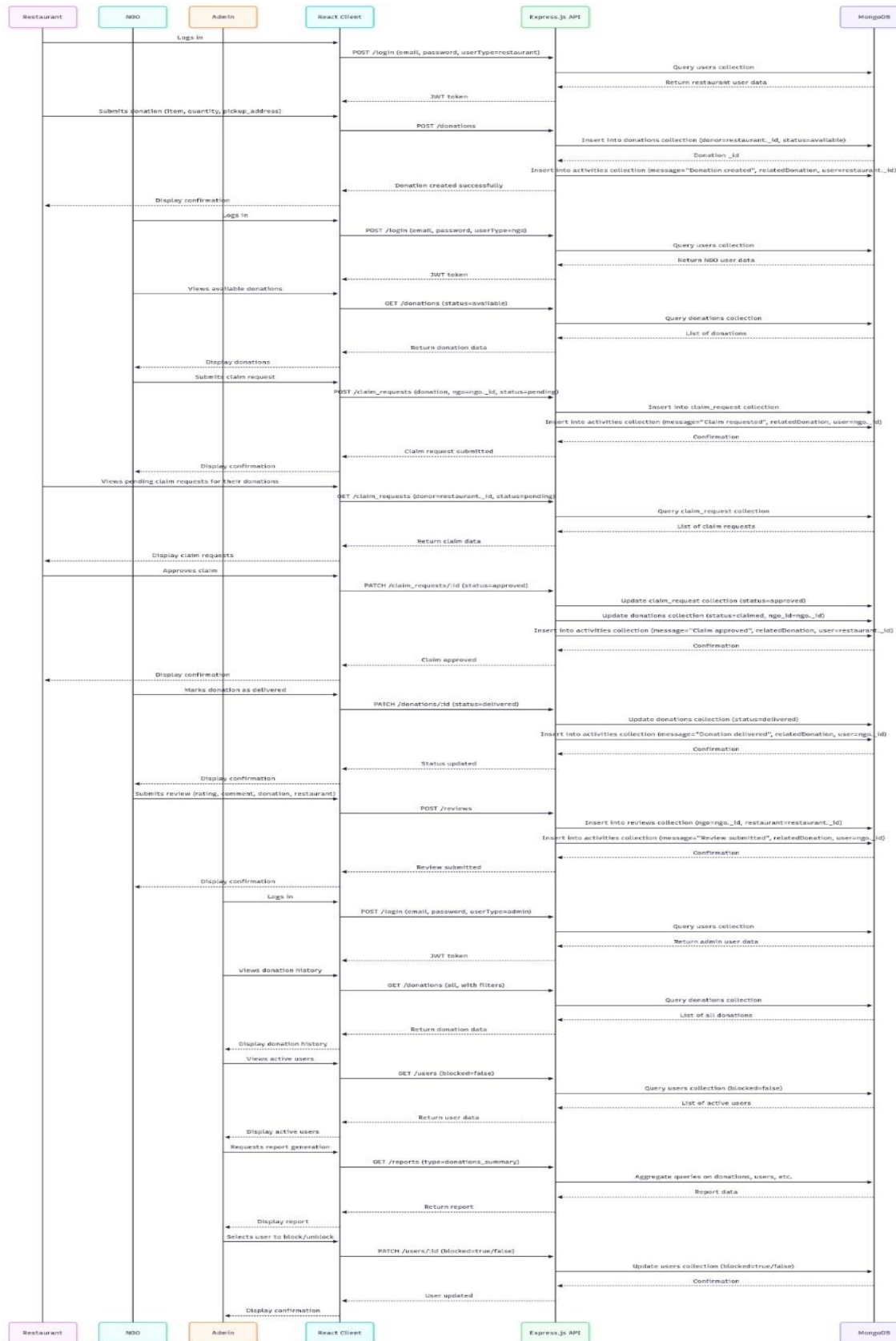
4.4.5 Review1

- A positive feedback given by NGO1: rating = 5, comment = *"Food was fresh and well packed!"*.
- Refers to Donation1 and acknowledges Restaurant1.

4.4.6 Activity1

- A system log message: *"Donation created by Restaurant"* at 2025-09-12 14:05.
- Belongs to Restaurant1 and is related to Donation1.

4.5 Sequence Diagram



Donation lifecycle in the food waste donation system ensures transparency, security, and efficient handling of donations:

4.5.1 Restaurant Authentication:

- The Restaurant logs in using credentials.
- The system validates credentials via **JWT authentication** and issues a secure token.

4.5.2 Donation Creation:

- Restaurant uses the dashboard to create a **donation** (food item, quantity, expiry time, pickup address).
- The backend validates the data and stores the donation in **MongoDB**.
- The donation status is initially set to **Available**.

4.5.3 NGO Authentication & Request:

- The NGO logs in, also authenticated via JWT.
- The NGO requests a list of available donations.
- The system queries MongoDB and returns all unclaimed donations.

4.5.4 Claiming a Donation:

- The NGO selects a donation and submits a **claim request**.
- The system updates the donation status to **Claimed/Pending Pickup**.

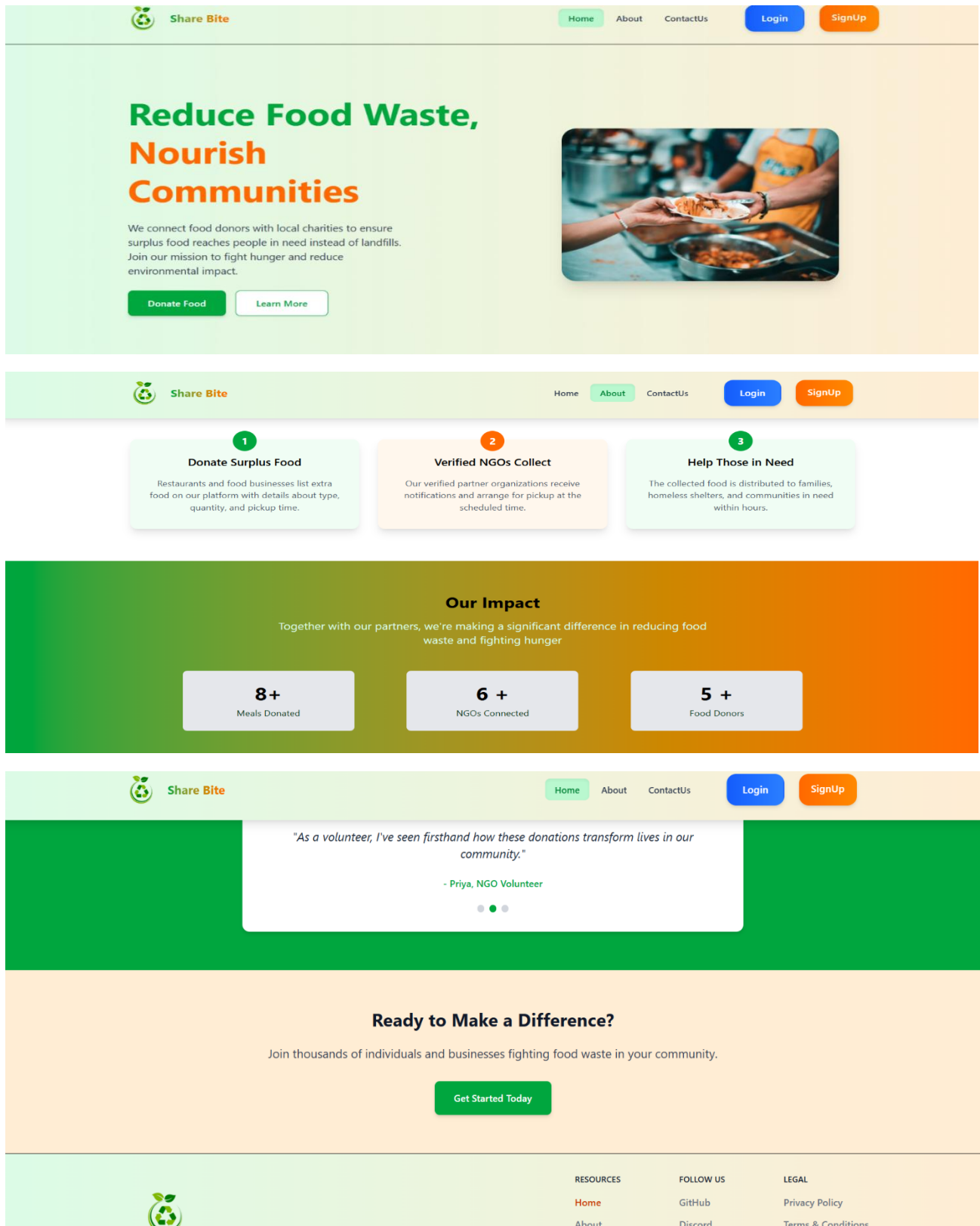
4.5.5 Admin Monitoring:

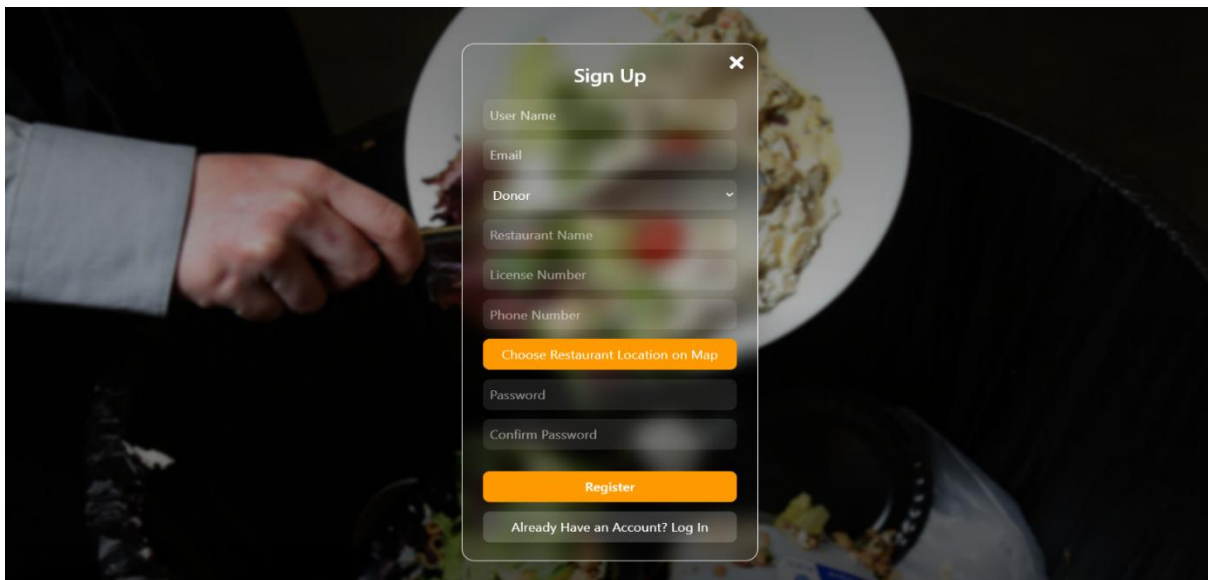
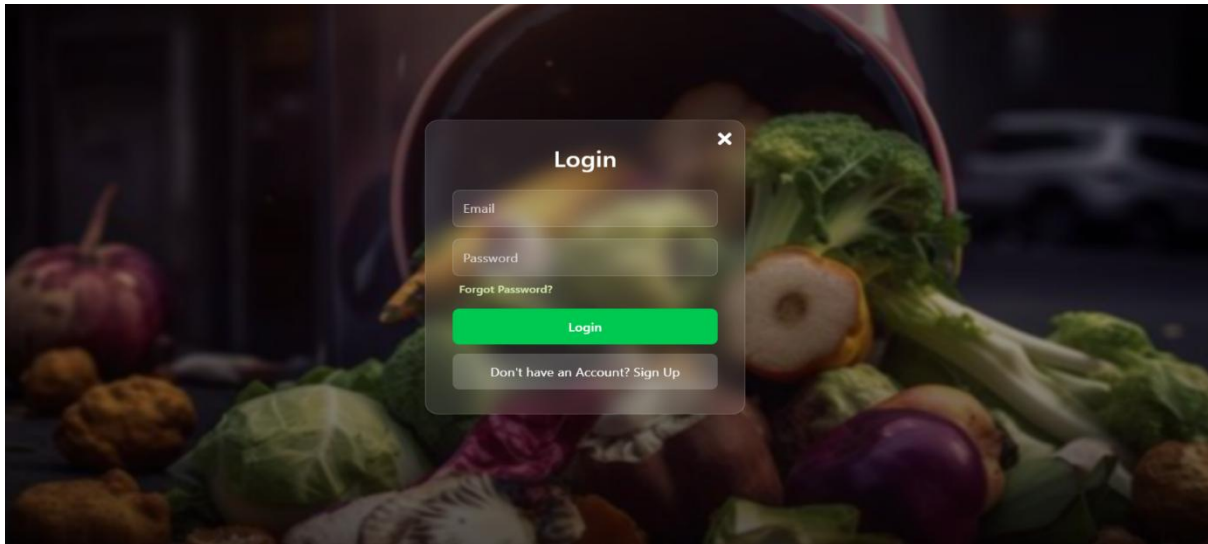
- The Admin dashboard provides a centralized view of users, donations, claims, and system activities.
- Admin ensures the donation process follows guidelines and resolves disputes if needed.

4.5.6 Completion:

- Once the NGO picks up the food, the donation is marked as **Completed**.
- NGO can leave feedback (Reviews).

5 Chapter 5: Graphical User Interfaces:





≡ NGO Dashboard 🔍 Available Donations 👤 My Claims 📌 Completed Donations 📍 Donation Map ⚙️ Rejected Donations ⚙️ Settings Logout	Completed Donations		
	🔍 Search donations...		
	Croissants, muffins, danishes Donor: Rauf Res Quantity: 40 Delivered: Invalid Date	Pasta dishes (lasagna, spaghetti) Donor: Rauf Res Quantity: 40 Delivered: Invalid Date	Soups & stews (kept at safe temps) Donor: pc hotel Quantity: 20 Delivered: Invalid Date
	Pasta dishes (lasagna, spaghetti) Donor: Ajwa Bakers Quantity: 30 Delivered: Invalid Date	Vegetable platters (carrots, celery, cucumbers) Donor: Ajwa Bakers Quantity: 10 Delivered: Invalid Date	Croissants, muffins, danishes Donor: Ajwa Bakers Quantity: 10 Delivered: Invalid Date
	Sandwiches/wraps (unwrapped, fresh) Donor: Ajwa Bakers Quantity: 14 Delivered: Invalid Date	Cookies & biscuits Donor: Ajwa Bakers Quantity: 40 Delivered: Invalid Date	Rice-based dishes (biryani, fried rice) Donor: Ajwa Bakers Quantity: 10 Delivered: Invalid Date

NGO Dashboard

Available Donations

My Claims

Completed Donations

Donation Map

Rejected Donations

Settings

Logout

Donation Map

NGO Dashboard

Available Donations

My Claims

Completed Donations

Donation Map

Rejected Donations

Settings

Rejected Donations

Search donations...

Rice-based dishes (biryani, fried rice)

Donor: Ajwa Bakers
Quantity: 10
Rejected On: 7/21/2025, 12:02:56 AM
Status: rejected

Fresh bread rolls/baguettes

Donor: Ajwa Bakers
Quantity: 10
Rejected On: 7/20/2025, 11:54:13 PM
Status: rejected

Pasta dishes (lasagna, spaghetti)

Donor: Ajwa Bakers
Quantity: 15
Rejected On: 7/20/2025, 11:54:21 PM
Status: rejected

Restaurant Dashboard

Dashboard
Food Donations
NGOs Nearby
Settings
Logout

Manage Food Donations

Search donations...

Add Donation

Soups & stews (kept at safe temps) delivered

Quantity: 10pieces
Peshawer
Delivered
Assigned to: Hope Bites

Cakes (whole or sliced, no cream if perishable) delivered

Quantity: 30pieces
Karachi
Delivered
Assigned to: Hope Bites

Sandwiches/wraps (unwrapped, fresh) delivered

Quantity: 10grams
Rawalpindi
Delivered
Assigned to: Hope Bites

Pasta dishes (lasagna, spaghetti) delivered

Quantity: 40boxes
Islamabad
Delivered
Assigned to: Eco Eats

Croissants, muffins, danishes delivered

Quantity: 40packets
Bahria Town Phase 4
Delivered
Assigned to: Eco Eats

Fresh bread rolls/baguettes expired

Quantity: 10packets
Rawalpindi
Expired

Restaurant Dashboard

- Dashboard
- Food Donations
- NGOs Nearby
- Settings

Nearby NGOs

Eco Eats 03495556578 7.99 km Contact	Food for all 03495556553 34.26 km Contact
Hope Bites 0349555658 28.84 km Contact	Hope Bites 03455456567 33.25 km Contact

Logout

Admin Dashboard

- Dashboard
- Manage Users
- Manage Donations
- Reports
- Map View

Dashboard Overview

Total Donations
37

Active Users
12

Pending Requests
2

Top Donors

1	Ajwa Bakers restaurant1@gmail.com	Total Donations 24
2	Rauf Res restaurant3@gmail.com	Total Donations 6
3	pc hotel restaurant2@gmail.com	Total Donations 6
4	Agha Jee restaurant6@gmail.com	Total Donations 1

Logout

Admin Dashboard

- Dashboard
- Manage Users
- Manage Donations
- Reports
- Map View

Manage Users

View and manage all system users

Filter by:

All Users
Restaurants
NGOs

Ngo1 ngo1@gmail.com Block User	Myres restaurant1@gmail.com ★★★★★ 0.0 / 5 Block User	Ngo2 ngo2@gmail.com Block User
Restaurant2 restaurant2@gmail.com ★★★★★ 0.0 / 5 Block User	ngo3 ngo3@gmail.com Block User	Ngo4 ngo4@gmail.com Block User
ngo5 Block User	Ngo6 Block User	Restaurant 3 Block User

Logout

Admin Dashboard

Dashboard
Manage Users
Manage Donations
Reports
Map View

Logout

Manage Donations

Filter by Status: All

Agha Jee
Donuts (day-old acceptable)
10 pieces
Rawat
8/26/2025
delivered

Rauf Res
Croissants, muffins, danishes
40 packets
Bahria Town Phase 4
8/26/2025
delivered

Rauf Res
Pasta dishes (lasagna, spaghetti)
40 boxes
Islamabad
8/26/2025
delivered

Rauf Res
Sandwiches/wraps (unwrapped, fresh)
10 grams
Rawalpindi
8/21/2025
delivered

Rauf Res
Cakes (whole or sliced, no cream if perishable)
30 pieces
Karachi
8/21/2025
delivered

Rauf Res
Soups & stews (kept at safe temps)
10 pieces
Peshawer
8/21/2025
delivered

Pr Hotel

Aiwa Bakers

Aiwa Bakers

Admin Dashboard

Dashboard
Manage Users
Manage Donations
Reports
Map View

Logout

Map View

NGOs
Restaurants

Admin Dashboard

Dashboard
Manage Users
Manage Donations
Reports
Map View

Logout

Reports and Analytics

Overall Donation Summary

Total Donations
37

Change from Last Month
-75%
Decrease from last month

Donation Status Distribution

Available	0 (0%)
Delivered	16 (43%)
Expired	21 (57%)

Donation Trends (Past 4 Months)

Average Donations: 9

Jun 2025
0

Jul 2025
27
+27 from previous month

Aug 2025
8
-19 from previous month

6 CHAPTER 6: Testing

6.1 Introduction

Testing is an essential phase of the software development life cycle to ensure that the developed system meets functional and non-functional requirements. In the Food Waste Donation System, testing was carried out to validate correctness, performance, security, and usability. Both **Black Box Testing** (functional testing from the user's perspective) and **White Box Testing** (verification of internal logic and APIs) were applied. The goal was to confirm that Restaurants, NGOs, and Admins could perform their tasks smoothly without errors.

6.2 Test Scenario

Test Case ID: UC-01

Test Case Name: User Registration

Actor: New User (Restaurant/NGO/Admin)

Test Priority: High

Preconditions:

- User is not already registered in the system.
- Registration page is accessible.

Postconditions:

- New account is created.
- User can log in with credentials.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	Launch Application	Vite + React	Registration page loads	Registration page loads	Chrome	PASS	Page Loaded
2	Enter valid details	Name, Email, Role etc	User registered	User registered	Chrome	PASS	Registration successful
3	Enter duplicate email	Email: ali@test.com	Error: Email exists	Error displayed	Chrome	PASS	Validation working

Test Case ID: UC-02

Test Case Name: Create Donation

Actor: Restaurant

Test Priority: High

Preconditions:

- User is logged in as Restaurant.

Postconditions:

- New donation created and visible to NGOs.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	Login	Restaurant credentials	Dashboard visible	Dashboard visible	Chrome	PASS	Login Successful
2	Open Create Donation form	-	Form displayed	Form displayed	Chrome	PASS	Form Loaded
3	Fill details	Food: Rice Qty: 10 plates Pickup: 6PM	Donation created	Donation created	Chrome	PASS	Donation Added

Test Case ID: UC-03

Test Case Name: Manage Donations

Actor: Restaurant

Test Priority: High

Preconditions:

- User is logged in as Restaurant.
- Has existing donations.

Postconditions:

- Donations updated or removed.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	View Donations	-	List visible	List visible	Chrome v118	PASS	View OK
2	Edit Donation	Change qty 10→15	Donation updated	Donation updated	Chrome v118	PASS	Edit OK
3	Delete Donation	Select donation	Donation deleted	Donation deleted	Chrome v118	PASS	Delete OK

Test Case ID: UC-04

Test Case Name: View Available Donations

Actor: NGO

Test Priority: High

Preconditions:

- User is logged in as NGO.

Postconditions:

- Available donations displayed.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	Login	NGO credentials	Dashboard visible	Dashboard visible	Chrome	PASS	Login OK
2	Browse Donations	Filter by location	Relevant donations displayed	Relevant donations displayed	Chrome	PASS	Filter OK

Test Case ID: UC-05

Test Case Name: Claim Donation

Actor: NGO

Test Priority: High

Preconditions:

- User is logged in as NGO.
- Donation is available.

Postconditions:

- Donation status changes to claimed.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	Select Donation	Donation #123	Donation details visible	Donation details visible	Chrome	PASS	Details OK
2	Click Claim	-	Donation claimed	Donation claimed	Chrome	PASS	Claim Successful

Test Case ID: UC-06

Test Case Name: Schedule Pickup

Actor: NGO

Test Priority: High

Preconditions:

- Donation claimed by NGO.

Postconditions:

- Pickup time scheduled & communicated.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	Open Claimed Donation	-	Donation details visible	Donation details visible	Chrome	PASS	OK
2	Schedule Pickup	Time: 7PM	Pickup confirmed	Pickup confirmed	Chrome	PASS	Pickup Scheduled

Test Case ID: UC-07

Test Case Name: Write Review

Actor: NGO

Test Priority: Medium

Preconditions:

- NGO has received donation.

Postconditions:

- Review added to Restaurant profile.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	Open Completed Donation	-	Donation details visible	Donation details visible	Chrome	PASS	OK
2	Submit Review	Rating: 5 Comment: Good	Review saved	Review saved	Chrome	PASS	Review Added

Test Case ID: UC-08

Test Case Name: View Reviews

Actor: NGO

Test Priority: Low

Preconditions:

- User is logged in as NGO.

Postconditions:

- Reviews displayed for Restaurants.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	Login	NGO credentials	Dashboard visible	Dashboard visible	Chrome	PASS	Login OK
2	Open Restaurant Reviews	Select Restaurant A	Reviews displayed	Reviews displayed	Chrome	PASS	Reviews OK

Test Case ID: UC-09**Test Case Name:** Manage Users**Actor:** Admin**Test Priority:** Medium**Preconditions:**

- User is logged in as Admin.

Postconditions:

- Users updated per admin actions.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	Open User Management	-	User list displayed	User list displayed	Chrome	PASS	List OK
2	Block User	User NGO123	User blocked	User blocked	Chrome	PASS	Block OK
3	Delete User	User Rest456	User deleted	User deleted	Chrome	PASS	Delete OK

Test Case ID: UC-10**Test Case Name:** View Analytics**Actor:** Admin**Test Priority:** Low**Preconditions:**

- User is logged in as Admin.

Postconditions:

- Analytics displayed.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	Open Analytics	-	Charts visible	Charts visible	Chrome	PASS	OK
2	Filter Reports	Date range	Filtered results displayed	Filtered results displayed	Chrome	PASS	Filter OK

Test Case ID: UC-11

Test Case Name: Confirm Delivery

Actor: Ngo

Test Priority: High

Preconditions:

- Delivery completed.

Postconditions:

- Donation status changed to delivered.

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test Result	Test Comments
1	Confirm Delivery	-	Status: Delivered	Status: Delivered	Chrome	PASS	Delivery Confirmed

6.3 Test Plan

The test plan describes the strategy and resources used during testing.

- **Scope:** Authentication, role-based access, donation lifecycle, reviews, claims, and admin functionalities.
- **Objectives:** Ensure correctness of donation flow, validate user roles, and confirm data integrity.
- **Resources:**
 - Development Environment: MERN Stack (MongoDB, Express.js, React, Node.js)
 - Testing Tools: Postman (API testing), Jest/Mocha (unit testing), Manual browser testing (Chrome v118)
- **Schedule:** Testing was performed incrementally after module implementation (e.g., user module, donation module, claim module).
- **Responsibilities:** Testing was conducted by the development team; results were documented for final verification.

6.4 Definition of Test Cases:

A test case is a detailed specification including preconditions, actions, expected results, and actual results. Each test case in this project was tied to a use case (UC) to ensure complete coverage.

6.5 Test Case Results:

6.5.1 Black Box Testing:

- All functional requirements were tested from the user's perspective.
- Restaurants successfully logged in, created, managed, and deleted donations.
- NGOs were able to view available donations, claim them, schedule pickups, and confirm deliveries.
- Admin was able to manage users, view donations, monitor claims, and analyze reports.
- Validation errors (e.g., duplicate emails, expired donations) worked as expected.

Result: All test cases passed, ensuring the system works according to requirements.

6.5.2 White Box Testing:

- **Controllers and APIs** were tested using Postman and unit tests.
- Authentication middleware successfully validated JWT tokens.
- Donation controller stored, updated, and retrieved donations correctly from MongoDB.
- Claim controller prevented expired claims and ensured only valid NGOs could claim.
- Review controller stored ratings and comments linked with correct donations and restaurants.

Result: Internal logic and database interactions were validated successfully.

6.6 Conclusion:

The testing phase confirmed that the Food Waste Donation System met its objectives. Both black box and white box testing demonstrated system stability, correct functionality, and data consistency. The donation lifecycle, from restaurant contribution to NGO claim and delivery confirmation, worked seamlessly. Minor bugs were fixed during testing, resulting in a fully functional and reliable system ready for deployment.

7 CHAPTER 7: Conclusion and Future Work

7.1 Conclusion:

The Food Waste Donation System was developed with the primary aim of connecting **restaurants and NGOs** to minimize food wastage and support underprivileged communities. Using the **MERN stack (MongoDB, Express.js, React.js, Node.js)**, the system provides role-based dashboards for **Admin, Restaurant, and NGO**, ensuring secure access through JWT authentication.

The system enables restaurants to donate surplus food, NGOs to claim donations, and Admins to monitor users, donations, and claims in real-time. Important features such as **donation lifecycle management, reviews, analytics, and activity tracking** were successfully implemented.

Through comprehensive testing (both black-box and white-box), the system was validated for functionality, usability, and reliability. The results confirmed that the platform fulfills its objectives by:

- Reducing food wastage.
- Ensuring timely distribution of food to NGOs.
- Providing transparency and accountability through reviews and monitoring.
- Offering scalability for future enhancements.

In conclusion, the project successfully addresses a **real-world social issue** using technology, promoting sustainability and community welfare.

7.2 Future Work

Although the system is fully functional, there are several opportunities for improvement and expansion in future versions:

1. Mobile Application Development

- Build Android and iOS apps for wider accessibility and real-time notifications.

2. Geolocation & Route Optimization

- Enhance the map feature with route optimization for faster pickup and delivery.

3. Integration with SMS/Email Notifications

- Notify NGOs about nearby donations instantly via SMS, WhatsApp, or email.

4. Payment/Reward System

- Introduce reward points or tax credit reports for restaurants that donate regularly.

5. Advanced Analytics Dashboard

- Provide AI-driven insights for Admins, NGOs, and Restaurants to measure impact, donor performance, and food wastage reduction.

References:

React. (n.d.). *Getting started – React documentation*. From <https://reactjs.org/docs/getting-started.html>

Node.js Foundation. (n.d.). *Node.js v20 documentation*. From <https://nodejs.org/en/docs/>

Express.js Foundation. (n.d.). *Express.js guide*. From <https://expressjs.com/>

MongoDB, Inc. (n.d.). *MongoDB manual*. From <https://www.mongodb.com/docs/>

JWT.io. (n.d.). *Introduction to JSON Web Tokens*. From <https://jwt.io/introduction>